

3.2 Procedure DNA isolation from water (eDNA dual filter capsule)

1. **Incubate the eDNA dual filter capsule containing the preservation liquid at 55 - 60 °C for 60 min to release all DNA from cell material.** Samples older than one day are already lysed at room temperature and can be used directly in step 4.
2. **Prepare 15 ml tubes with 750 µl S2 (Phase separation solution).** Prepare as much tubes as the amount of samples are present, prepare also one additional tube. This additional tube will be the internal negative control (INC). Preparing these tube before handling the samples will prevent contamination of liquids present in the kit.
3. **Prepare 2 ml tubes with 100 µl S3(Precipitation buffer) and 900 µl S4 (Precipitation solution).** Prepare as much tubes as the amount of samples are present, prepare also one additional tube. This additional tube will be the internal negative control (INC). Preparing these tube before handling the samples will prevent contamination of liquids present in the kit.

Note: Steps 4 to 9 is clarified in a video tutorial on:

<https://sylphium.com/webshop/product/syl002>



4. **Remove the cap of the outlet of the filter capsule and connect the 60 ml syringe (used during sampling) on the outlet.**
5. **Keep the filter capsule above the syringe, remove the second cap and pull the plunger slowly until the liquid is inside the syringe.** Take care not to spill any of the DNA containing liquid. This can be a source of DNA contamination for other samples. If any spill occur, clean al contaminated material with DNA decontamination reagent
6. **To maximize liquid extraction from the filter capsule, push and pull the plunger slowly a few times (air in the capsule, liquid out).** Be aware not to push the liquid back into the filter capsule.
7. **Add the liquid inside the syringe to the 15 ml tube containing 750 µl S2 (Phase separation solution). Number this tube with the number written on the sample bag.**
8. **Take the additional 15 ml tube and add 1 ml S1 (Internal negative control) into this tube, mix by shaking and mark this tube as INC.** This is the Internal Negative Control and it should be handled in the coming steps like the other samples.

9. **Centrifuge samples at +/- 4.000 x g for 5 minutes.** A bilayer will be formed. The top layer is the water phase and contains the DNA. Between the two liquid layers a small solid layer with denaturated proteins can be formed.
10. **Transfer max 1000 µl of the water layer to the prepared mixture (S3 and S4) and mix well by gently shaking and cool down to -20 °C for at least 20 min.** In this step DNA will precipitate while salts and other unwanted substances will stay in solution. Prevent transfer of the protein and CIA layer to the new tube. Is a metal cooling block is placed in the freezer, this step can be shorter.
11. **Transfer the remaining water layer (+/- 1000 µl) to an empty 2 ml tube and use this as a backup. Store this tube in a -20 °C freezer.** If no backup is required, these samples can be used in extraction to maximize eDNA recovery from the sample. Make additional tubes as described in step 2 if these samples will be isolated as well.
12. **Centrifuge samples for 30-45 minutes (for small eDNA fragments) or 5 minutes for genomic DNA (bacteria, etc.) at maximum speed (≥13.000 g). Remove supernatant by decanting and pipetting.** Be careful, the DNA pellet is not firmly attached to the tube. It's important to remove all liquid to increase sensitivity of qPCR detecting.
13. **Add 500 µl S5 (Wash solution), mix and centrifuge samples at maximum speed (≥13.000 g) for 5 minutes. Remove supernatant completely by pipetting and remove all liquid. Repeat this step to remove PCR inhibiting substances.** Be careful, the DNA pellet is still not firmly attached to the tube.
14. **Dissolve the pellet in 100 µl S6 (eDNA conservation buffer) and use 2 µl for qPCR.**